Electricity Markets as Games: Bounding the Price of Anarchy

1 The Electricity Market Game

1.1 Market model

We consider a finite set $\mathcal{I} = \{1, \dots, n\}$ of generators that together have to cover a fixed demand $D > 0$. Generator i produces the amount $x_i \in \mathbb{R}$ of energy at the *true* cost

$$C_i(x_i) = a_i x_i^2,$$

where the coefficient $a_i > 0$ is private information of generator i. The only constraint on the joint production $x = (x_i)_{i \in \mathcal{I}}$ is that supply equals demand, $\sum_{i \in \mathcal{I}} x_i = D$. Everything that we discuss here stays true for market models in which additionally network and capacity constraints are considered. It is only for simplicity that we stay with this simple market model.

Social optimum. The socially optimal dispatch covers the demand at minimum true cost c^*:

$$c^* := \min_x \sum_{i \in \mathcal{I}} a_i x_i^2 \quad \text{s.t.} \quad \sum_{i \in \mathcal{I}} x_i = D. \tag{1}$$

Market clearing. The operator does not know the true coefficients a_i. Instead, every generator *declares* (bids) a coefficient

$$\tilde{a}_i \in \mathcal{H}_i := [10^{-3}, a_{\max}],$$

which may or may not be equal to its true coefficient a_i. We assume that the true coefficients lie in the same interval, $a_i \in \mathcal{H}_i$. We write $\tilde{a} = (\tilde{a}_i)_{i \in \mathcal{I}}$ for the bid profile and $\mathcal{H} := \mathcal{H}_1 \times \cdots \times \mathcal{H}_n$ for the joint bid space. The lower bound 10^{-3} keeps all bids strictly positive, so that the market-clearing problem below always has a unique solution.

Given the bids, the operator solves the *market-clearing problem*

$$\min_x \sum_{i \in \mathcal{I}} \tilde{a}_i x_i^2 \quad \text{s.t.} \quad \sum_{i \in \mathcal{I}} x_i = D \qquad \text{(dual variable } \gamma\text{)}, \tag{2}$$

that is, it minimizes the system cost as if the bids were truthful. The solution $x^*(\tilde{a})$ is the *dispatch*, and the optimal dual variable $\gamma(\tilde{a})$ of the balance constraint is the *market-clearing price* [1]. Every generator is paid $\gamma(\tilde{a})$ per unit of energy it is dispatched.

KKT formulation of the market clearing. Problem (2) is a strictly convex quadratic program with a single linear equality constraint. Hence, the KKT conditions are necessary and sufficient for optimality, and the market-clearing problem can be expressed equally via its KKT conditions: a pair (x^*, γ) is the dispatch and price for the bids $\tilde{a}$ if and only if

$$\begin{aligned} 2\,\tilde{a}_i\, x_i^* - \gamma = 0 \qquad & \forall\, i \in \mathcal{I}, && \text{(stationarity)} \\ \sum_{i \in \mathcal{I}} x_i^* = D. \qquad & && \text{(primal feasibility)} \end{aligned} \tag{3}$$

The stationarity condition has a clear economic meaning: every generator is dispatched up to the point where its *declared* marginal cost $2\tilde{a}_i x_i$ equals the price.

We will need this KKT formulation later on, when we optimize over expressions that depend on the outcome of the market clearing (Section 3). There, instead of treating $x^*(\tilde{a})$ and $\gamma(\tilde{a})$ as the result of an inner optimization problem, we add x^* and γ as additional variables and impose (3) as constraints.

Profits and social cost. The profit of generator i is its revenue minus its *true* production cost,

$$\pi_i(\tilde{a}) = \gamma(\tilde{a})\, x_i^*(\tilde{a}) - a_i\, x_i^*(\tilde{a})^2.$$

The social cost of a bid profile is the true cost of the resulting dispatch,

$$c(\tilde{a}) = \sum_{i \in \mathcal{I}} a_i\, x_i^*(\tilde{a})^2.$$

For truthful bids, $\tilde{a} = a$, problem (2) coincides with (1), and hence $c(a) = c^*$.

Remark 1 (Quadratic-plus-linear costs). Everything in these notes also works for costs of the form $C_i(x_i) = a_i x_i^2 + b_i x_i$, where generators bid a pair $(\tilde{a}_i, \tilde{b}_i)$ from a box $\mathcal{H}_i \subset \mathbb{R}^2$, where again the quadratic coefficient is strictly greater than 0. The only property we really use (in Lemma 1) is that the cost is *linear in its parameters*. The stationarity condition in (3) then becomes $2\tilde{a}_i x_i + \tilde{b}_i - \gamma = 0$. If, in addition, non-negativity constraints $x_i \geq 0$ become relevant, the KKT conditions also contain the corresponding multipliers and complementarity conditions.

1.2 The game

Each generator chooses its bid to maximize its own profit. This defines the game G:

- **Players:** generators $i \in \mathcal{I}$.
- **Strategies:** bids $\tilde{a}_i \in \mathcal{H}_i$.
- **Loss functions:** negative profits $J_i(\tilde{a}) = -\pi_i(\tilde{a})$.
- **Social cost:** $c(\tilde{a}) = \sum_{i \in \mathcal{I}} a_i\, x_i^*(\tilde{a})^2$.

Note that the loss of player i depends on the bids of *all* players, because both the price γ and the dispatch x_i^* result from the market-clearing problem. We write $\tilde{a} = (\tilde{a}_i, \tilde{a}_{-i})$, where $\tilde{a}_{-i}$ collects the bids of all players except i, and $J_i(\tilde{a}_i', \tilde{a}_{-i})$ denotes the loss of player i when it alone switches to the bid $\tilde{a}_i'$.

1.3 Equilibria and the Price of Anarchy

As solution concept, we use the *Coarse Correlated Equilibrium* (CCE). A CCE is a probability distribution q over bid profiles $\tilde{a} \in \mathcal{H}$ such that no player can lower its expected loss by committing to a fixed bid:

$$\mathbb{E}_{\tilde{a} \sim q}\big[J_i(\tilde{a})\big] \;\leq\; \mathbb{E}_{\tilde{a} \sim q}\big[J_i(\tilde{a}_i', \tilde{a}_{-i})\big] \qquad \forall\, \tilde{a}_i' \in \mathcal{H}_i,\ \forall\, i \in \mathcal{I}. \tag{4}$$

We denote the set of all CCE by CCE. Two reasons for this choice: (i) the market is cleared repeatedly, and if every generator adapts its bid with a no-regret learning algorithm, the empirical

distribution of bids converges to the set of CCE [2]; (ii) every Nash and every Correlated Equilibrium is a CCE, so all bounds below also hold for these equilibria.

The expected social cost at an equilibrium $q \in \mathrm{CCE}$ is

$$c^{\mathrm{eq}}(q) = \mathbb{E}_{\tilde{a}\sim q}\big[c(\tilde{a})\big],$$

and the *Price of Anarchy* (PoA) [3] is the worst-case ratio between equilibrium cost and optimal cost:

$$\rho(G) = \frac{\sup_{q\in\mathrm{CCE}} c^{\mathrm{eq}}(q)}{c^*} \;\geq\; 1.$$

Our goal is to compute a certified *upper bound* on $\rho(G)$.

The main difficulty is the set CCE: it is a set of probability distributions over the continuum $\mathcal{H}$, defined by one inequality per player and per possible deviation $\tilde{a}'_i$. Maximizing over this infinite-dimensional set directly is hopeless. We therefore use a technique that bounds the PoA *without ever computing the equilibria*: the smoothness framework.

2 Generalized Smoothness

We use the generalized smoothness framework of Chandan et al. [4], which allows for a social cost c that is not related to the players' losses (in our game, c is the true production cost, not the sum of losses).

Proposition 1 (Chandan et al. [4], Theorem 2)**.** *If there exist $\lambda \geq 0$ and $\mu < 1$ such that*

$$\sum_{i\in\mathcal{I}} \Big[J_i(\tilde{a}'_i, \tilde{a}_{-i}) - J_i(\tilde{a})\Big] + c(\tilde{a}) \;\leq\; \lambda\, c(\tilde{a}') + \mu\, c(\tilde{a}) \qquad \forall\,(\tilde{a}, \tilde{a}') \in \mathcal{H}^2, \tag{5}$$

then $\rho(G) \leq \frac{\lambda}{1-\mu}$.

The proof is short and important for what follows.

Proof. Since (5) holds for all $\tilde{a}'$, it holds in particular for a bid profile $\tilde{a}'$ that is socially optimal, $c(\tilde{a}') = c^*$ (e.g., $\tilde{a}' = a$). Let $q \in \mathrm{CCE}$. Then

$$\begin{aligned} c^{\mathrm{eq}}(q) = \mathbb{E}_{\tilde{a}\sim q}\big[c(\tilde{a})\big] &\leq \mathbb{E}_{\tilde{a}\sim q}\Big[\sum_{i\in\mathcal{I}} \big[J_i(\tilde{a}'_i, \tilde{a}_{-i}) - J_i(\tilde{a})\big] + c(\tilde{a})\Big] \\ &\leq \mathbb{E}_{\tilde{a}\sim q}\big[\lambda\, c(\tilde{a}') + \mu\, c(\tilde{a})\big] = \lambda\, c^* + \mu\, c^{\mathrm{eq}}(q). \end{aligned}$$

The first inequality holds because, by the CCE condition (4), each term $\mathbb{E}_{\tilde{a}\sim q}[J_i(\tilde{a}'_i, \tilde{a}_{-i}) - J_i(\tilde{a})]$ is non-negative. The second inequality is (5). Rearranging gives $c^{\mathrm{eq}}(q)/c^* \leq \lambda/(1-\mu)$ for every $q \in \mathrm{CCE}$. ∎

The best bound obtainable from Proposition 1 is

$$\rho^{\mathrm{gen}}(G) := \inf_{\lambda\geq 0,\ \mu<1} \frac{\lambda}{1-\mu} \quad \text{s.t.} \quad (5). \tag{6}$$

Using only optimal deviations. The proof only used (5) for *one* socially optimal profile $\tilde{a}'$. Let

$$\mathcal{H}^* := \left\{\tilde{a} \in \mathcal{H} : c(\tilde{a}) = c^*\right\}$$

be the set of bid profiles that lead to a socially optimal dispatch. Then we may also choose the optimal profile that gives the best bound:

$$\begin{aligned} \widehat{\rho}^{\text{gen}}(G) := \inf_{\lambda \geq 0,\ \mu < 1,\ \tilde{a}^* \in \mathcal{H}^*} \ & \frac{\lambda}{1-\mu} \\ \text{s.t.} \quad & \sum_{i \in \mathcal{I}} \Big[J_i(\tilde{a}_i^*, \tilde{a}_{-i}) - J_i(\tilde{a}) \Big] + c(\tilde{a}) \leq \lambda\, c^* + \mu\, c(\tilde{a}) \qquad \forall\, \tilde{a} \in \mathcal{H}. \end{aligned} \tag{7}$$

Since (7) has fewer constraints than (6) and an additional degree of freedom, $\widehat{\rho}^{\text{gen}}(G) \leq \rho^{\text{gen}}(G)$.

Remark 2 (Why (7) is hard). Problem (7) has one constraint for every $\tilde{a} \in \mathcal{H}$, i.e., uncountably many: it is a *semi-infinite program*. (For finite action sets, as in [4], there would be finitely many constraints.) Moreover, in general we know nothing about the structure of $\mathcal{H}^*$; it may be non-convex or even disconnected. Even finding a feasible point is therefore difficult. In the next section, we exploit the structure of the electricity market to get around this.

3 Computing a PoA Bound

3.1 Scaled truthful bids are socially optimal

The key observation is that the market outcome stays socially optimal if *all* generators scale their true coefficients by the same factor.

Lemma 1. *Let $\omega > 0$ be such that $\omega a \in \mathcal{H}$. Then $\omega a \in \mathcal{H}^*$. More precisely,*

$$x^*(\omega a) = x^*(a), \qquad \gamma(\omega a) = \omega\, \gamma(a).$$

Proof. With bids $\tilde{a} = \omega a$, the objective of (2) is $\omega \sum_i a_i x_i^2$, i.e., the objective of (1) multiplied by $\omega > 0$. Scaling the objective does not change the minimizer, so the dispatch is socially optimal, and it scales the dual variable by ω. (Alternatively, via the KKT conditions (3): if (x^*, γ) satisfies (3) for the bids a, then $(x^*, \omega\gamma)$ satisfies (3) for the bids ωa.) ■

The proof only uses that the cost is linear in its parameter; this is why the approach also works in the setting of Remark 1. Define the interval of admissible scaling factors

$$\Omega := \{\omega > 0 : \omega a \in \mathcal{H}\} = \left[\frac{10^{-3}}{\min_i a_i},\ \frac{a_{\max}}{\max_i a_i} \right],$$

which contains $\omega = 1$ (truthful bidding), and the corresponding set of optimal profiles $\mathcal{H}^*_\Omega := \{\omega a : \omega \in \Omega\} \subseteq \mathcal{H}^*$.

Remark 3 (How much do we lose?). In this simple market we actually have $\mathcal{H}^*_\Omega = \mathcal{H}^*$: since (1) has a unique minimizer, $c(\tilde{a}) = c^*$ requires $x^*(\tilde{a}) = x^*(a)$. By stationarity in (3), all x_i^* have the same sign as the price, and since they sum to $D > 0$, they are all positive. Stationarity for $\tilde{a}$ and for a then gives $\tilde{a}_i / a_i = \gamma(\tilde{a})/\gamma(a)$ for all i, i.e., $\tilde{a}$ is a multiple of a. In richer market models (e.g., with capacity or network constraints), $\mathcal{H}^*_\Omega$ is in general a proper subset of $\mathcal{H}^*$. For instance, a generator that produces at its capacity limit can raise its bid somewhat without changing the dispatch.

Restricting the optimal deviation $\tilde{a}^*$ in (7) to $\mathcal{H}^*_\Omega$ gives the following smoothness condition, tailored to electricity markets. Here and below, $J_i(\omega a_i, \tilde{a}_{-i})$ is the loss of player i when it alone deviates to the scaled truthful bid ωa_i.

Corollary 1 (Electricity market smoothness)**.** *If there exist* $\lambda \geq 0$, $\mu < 1$ *and* $\omega \in \Omega$ *such that*

$$\sum_{i \in \mathcal{I}} \Big[J_i(\omega a_i, \tilde{a}_{-i}) - J_i(\tilde{a}) \Big] + c(\tilde{a}) \;\leq\; \lambda\, c^* + \mu\, c(\tilde{a}) \qquad \forall\, \tilde{a} \in \mathcal{H}, \tag{8}$$

then $\rho(G) \leq \frac{\lambda}{1-\mu}$.

Proof. Repeat the proof of Proposition 1 with $\tilde{a}' = \omega a$, which is socially optimal by Lemma 1. ■

The best bound obtainable from Corollary 1 is

$$\rho^{\mathrm{EM}}(G) := \inf_{\lambda \geq 0,\ \mu < 1,\ \omega \in \Omega} \frac{\lambda}{1-\mu} \quad \text{s.t.} \quad (8). \tag{9}$$

Compared to (7), we now search over a *one-dimensional interval* Ω instead of the unknown set $\mathcal{H}^*$. In general, this is a restriction, so $\rho^{\mathrm{EM}}(G) \geq \widehat{\rho}^{\mathrm{gen}}(G)$ (with equality in our simple market). Problem (9) still has infinitely many constraints and is non-convex. We deal with both issues next.

3.2 Reformulation

To shorten notation, define for every $\omega \in \Omega$ and $\tilde{a} \in \mathcal{H}$ the sum of loss changes when every player unilaterally deviates to its scaled truthful bid,

$$\Delta_\omega(\tilde{a}) := \sum_{i \in \mathcal{I}} \Big[J_i(\omega a_i, \tilde{a}_{-i}) - J_i(\tilde{a}) \Big].$$

Lemma 2. *The optimization problem*

$$\inf_{\rho \geq 1,\ \nu \geq 0,\ \omega \in \Omega} \rho \quad \textit{s.t.} \quad \nu\, \Delta_\omega(\tilde{a}) + c(\tilde{a}) \leq \rho\, c^* \qquad \forall\, \tilde{a} \in \mathcal{H} \tag{10}$$

has the optimal value $\rho^{\mathrm{EM}}(G)$.

Proof. Multiply (8) by $\nu := \frac{1}{1-\mu} > 0$ and set $\rho := \frac{\lambda}{1-\mu} \geq 0$. Since $\nu(1-\mu) = 1$, the constraint becomes $\nu\, \Delta_\omega(\tilde{a}) + c(\tilde{a}) \leq \rho\, c^*$, and the objective becomes ρ. Two further changes do not affect the optimal value:

- *Adding* $\rho \geq 1$*:* every PoA bound is at least 1.
- *Allowing* $\nu = 0$*:* if $(\rho, 0, \omega)$ is feasible, then $c(\tilde{a}) \leq \rho c^*$ for all $\tilde{a}$. With $M := \max\{0, \max_{\tilde{a} \in \mathcal{H}} \Delta_\omega(\tilde{a})\} < \infty$, the point $(\rho + \varepsilon M/c^*,\ \varepsilon,\ \omega)$ is feasible for every $\varepsilon > 0$, and its objective converges to ρ as $\varepsilon \to 0$.

■

Problem (10) has a very convenient structure:

- For fixed ω, it is a *linear* program in only two variables (ρ, ν) (with infinitely many constraints). The only non-linearity comes from the scalar ω.
- The objective is the single variable ρ, which enters every constraint with the same coefficient c^*. Hence, if a candidate (ρ, ν, ω) violates every constraint by at most ε, then $(\rho + \varepsilon/c^*, \nu, \omega)$ is feasible. An approximately feasible point is turned into a feasible one by a simple shift.

Algorithm 1: Computing a PoA bound for the electricity market game

Input : True coefficients a, demand D, bound $a_{\max}$, tolerances $\varepsilon > 0$ and $\varepsilon_{\text{gss}} > 0$, finite initial set $\mathcal{H}' \subset \mathcal{H}$ (e.g., random bid profiles)

Output: PoA bound $\rho_{\varepsilon}^{\text{Alg1}}(G)$

$k \leftarrow 0$

while *true* **do**

  $k \leftarrow k + 1$

  `/* Inner loop: solve the relaxation over` $\mathcal{H}'$ `*/`

  Run a golden-section search over $\omega \in \Omega$ (tolerance ε_{gss}) to minimize $\varphi_{\mathcal{H}'}(\omega)$, solving the LP (11) at each evaluated ω. Let $(\rho_k, \nu_k, \omega_k)$ be the best point found.

  `/* Separation: find the most violated constraint */`

  $\tilde{a}' \leftarrow \arg\max_{\tilde{a} \in \mathcal{H}} g(\tilde{a}; \rho_k, \nu_k, \omega_k)$ and $V_k \leftarrow g(\tilde{a}'; \rho_k, \nu_k, \omega_k)$

  if $V_k \leq \varepsilon$ **then**

    return $\rho_{\varepsilon}^{\text{Alg1}}(G) := \rho_k + \varepsilon / c^*$

  else

    $\mathcal{H}' \leftarrow \mathcal{H}' \cup \{\tilde{a}'\}$

3.3 The algorithm

The two observations above lead to the two ideas of our approach.

1. **Infinitely many constraints ⇒ cutting planes.** Keep a finite set $\mathcal{H}' \subset \mathcal{H}$ of constraints, solve the problem only for these, then search $\mathcal{H}$ for the most violated constraint and add it to $\mathcal{H}'$. Repeat.

2. **Non-linearity in ω ⇒ one-dimensional search.** In the spirit of Benders decomposition [5], treat ω as the complicating variable: once it is fixed, what remains is a small LP in (ρ, ν). Since ω is a scalar, we search over it with a golden-section search.

For a fixed ω and a finite set $\mathcal{H}'$, let $\varphi_{\mathcal{H}'}(\omega)$ denote the optimal value of the finite LP

$$\varphi_{\mathcal{H}'}(\omega) := \min_{\rho \geq 1,\ \nu \geq 0} \ \rho \quad \text{s.t.} \quad \nu\, \Delta_\omega(\tilde{a}) + c(\tilde{a}) \leq \rho\, c^* \qquad \forall\, \tilde{a} \in \mathcal{H}'. \tag{11}$$

For a candidate (ρ, ν, ω), the *violation* of the constraint belonging to $\tilde{a} \in \mathcal{H}$ is

$$g(\tilde{a}; \rho, \nu, \omega) := \nu\, \Delta_\omega(\tilde{a}) + c(\tilde{a}) - \rho\, c^*.$$

The candidate is feasible for (10) if and only if $\max_{\tilde{a} \in \mathcal{H}} g(\tilde{a}; \rho, \nu, \omega) \leq 0$.

Algorithm 1 alternates between two steps. In the *inner loop*, it solves the relaxation of (10) that only contains the constraints in the finite set $\mathcal{H}'$: a golden-section search over ω, where each evaluation is the two-variable LP (11). In the *separation step*, it looks for the bid profile $\tilde{a}'$ whose constraint is violated most by the current candidate. If the violation exceeds ε, then $\tilde{a}'$ is added to $\mathcal{H}'$ and the relaxation is solved again. Otherwise, the algorithm stops.

Two points are worth emphasizing:

- *The returned value is a valid bound.* At termination, $(\rho_k, \nu_k, \omega_k)$ violates every constraint by at most ε. By the shift argument above, $(\rho_k + \varepsilon/c^*, \nu_k, \omega_k)$ is feasible for (10), and therefore (by Lemma 2 and Corollary 1) $\rho(G) \leq \rho_k + \varepsilon/c^*$. So the algorithm uses only ε-feasibility, for which cutting plane methods have good convergence results [6], but still returns a certified bound.

- *The golden-section search need not be globally optimal.* If $\varphi_{\mathcal{H}'}$ is not unimodal, the search may only find a local minimum. This affects the tightness of the bound, but not its validity, since Corollary 1 holds for *every* $\omega \in \Omega$.

3.4 Convergence

Assumption 1. *For every candidate* (ρ, ν, ω)*, the maximization* $\max_{\tilde{a} \in \mathcal{H}} g(\tilde{a}; \rho, \nu, \omega)$ *in the separation step can be solved in finite time.*

The maximum exists, since $\mathcal{H}$ is a compact box and g is continuous in $\tilde{a}$. This is the standard exact separation oracle assumption of cutting plane methods for semi-infinite programs [7].

This is where the KKT formulation (3) comes in. The function g depends on $\tilde{a}$ only through the outcomes of $n+1$ market clearings: the one for $\tilde{a}$ and, for every $i \in \mathcal{I}$, the one for $(\omega a_i, \tilde{a}_{-i})$. Introducing the dispatch and price of each of these market clearings as additional variables and imposing one copy of (3) for each of them, the separation problem becomes a single (finite-dimensional) optimization problem. Solving it is the computationally demanding part of the algorithm, since it is in general non-convex (e.g., the stationarity conditions are bilinear in the bids and the dispatch). However, Assumption 1 is stronger than needed:

- For the *validity* of the bound, it suffices to have a certified *upper bound* on the maximal violation; the algorithm may stop as soon as this upper bound is below ε.
- For *convergence*, it suffices to find a sufficiently violated constraint rather than the most violated one; cutting plane methods with such inexact oracles are analyzed in [8].

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